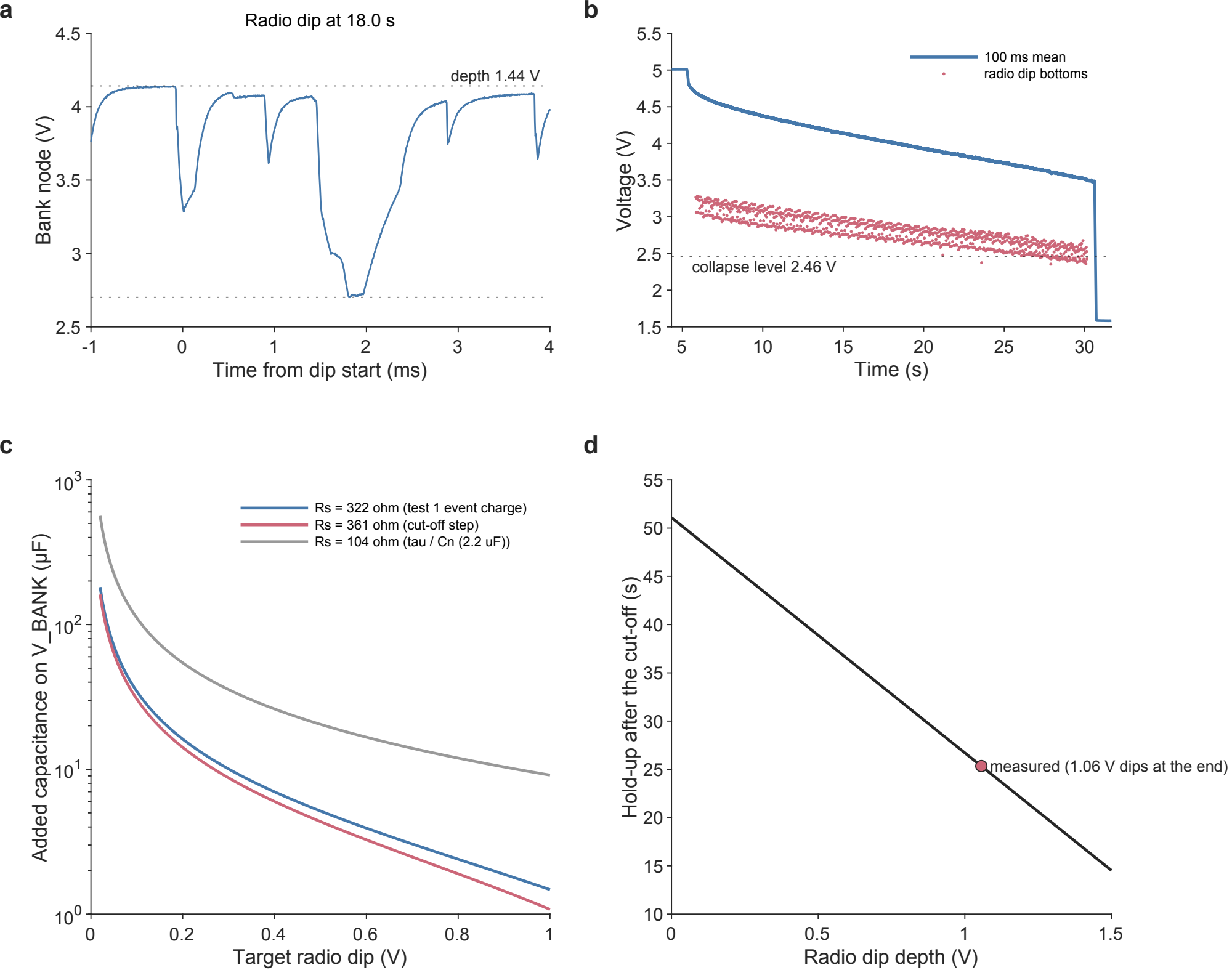

a
Radio dip at 18.0 s
depth 1.44 V
Bank node (V)
Time from dip start (ms)
b
100 ms mean
radio dip bottoms
collapse level 2.46 V
Voltage (V)
Time (s)
c
Rs = 322 ohm (test 1 event charge)
Rs = 361 ohm (cut-off step)
Rs = 104 ohm (tau / Cn (2.2 uF))
Added capacitance on V_BANK (µF)
Target radio dip (V)
d
measured (1.06 V dips at the end)
Hold-up after the cut-off (s)
Radio dip depth (V)

Decoupling capacitor for the radio dips during the capacitor-bank hold-up

Recording: DC_5.0V_CapBank_Test_4.bin (5 V DC supply cut by hand; scope at 200 kHz on the bank node).

1. Measured (bank only, after the cut-off)
   Hold-up 25.3 s; mean discharge -0.0442 V/s (0.73 mA at the nominal 16.5 mF).
   Radio events every 22.5 ms: dip 1.34 V deep, 0.83 ms wide (FWHM), recovery tau 0.230 ms,
   area A = 1.182 mV*s. Mean voltage step at the cut-off: 0.263 V.
   1 ms ticks: about 0.45 V for about 40 us (small charge; the same capacitor removes them).

2. Series resistance Rs between the bank and the board, and the charge per radio event
   The bank current is (dip / Rs), so the charge of one event is Q = A / Rs.
   (a) test 1 shunt, idle radio event charge 3.67 uC (direct): Rs = A / Q = 322 ohm.
   (b) cut-off step / mean current = 0.263 V / 0.73 mA: Rs = 361 ohm (Q = 3.28 uC).
   (c) recovery tau / nominal C54 = 0.230 ms / 2.2 uF: Rs = 104 ohm (Q = 11.33 uC).
   (a) and (b) agree. (c) is low because C54 (2.2 uF ceramic) has lost capacitance at 4-5 V bias:
   (a) implies Cn = tau / Rs = 0.71 uF. Burst current dip / Rs = 4.2 mA (a).

3. Capacitor on V_BANK (next to C54 at the LDO input, on the board side of Rs)
   The capacitor supplies each burst and recharges through Rs between events:
   dip = Q / (Cn + Cadd), so Cadd = Q / target - Cn (effective capacitance, at the bias voltage).
   target 0.05 V: Cadd = 71 uF from (a) [63 uF from (b), 224 uF from (c)]; recovery tau 23.6 ms
   target 0.10 V: Cadd = 35 uF from (a) [31 uF from (b), 111 uF from (c)]; recovery tau 11.8 ms
   target 0.20 V: Cadd = 16 uF from (a) [14 uF from (b), 54 uF from (c)]; recovery tau 5.9 ms
   target 0.30 V: Cadd = 10 uF from (a) [9 uF from (b), 36 uF from (c)]; recovery tau 3.9 ms
   target 0.50 V: Cadd = 5 uF from (a) [4 uF from (b), 20 uF from (c)]; recovery tau 2.4 ms
   Recovery stays well inside the 22.5 ms between radio events.

4. Hold-up gain
   The device collapsed when the dip bottoms reached 2.46 V, with the mean still at 3.52 V.
   With 0.10 V dips it would run until the mean reaches 2.56 V: about 23 s more (49 s instead of 25 s),
   at the end-of-discharge slope -0.0410 V/s (linear extrapolation; the current may change in LDO dropout).

5. Conclusion
   Each radio event draws about 3.7 uC (about 4 mA for 0.8 ms) through about 322 ohm between the bank
   and the board, so V_BANK dips by 1.3 V. About 35 uF of effective capacitance on V_BANK cuts the dips
   to 0.1 V (16 uF for 0.2 V) and adds roughly 23 s of hold-up at idle (about 92% longer).
   Ceramics lose 50-70% of their capacitance at 4-5 V: use X5R/X7R rated >= 10 V with 2-3x the effective
   value as nominal (e.g. 2 x 47 uF or 1 x 100 uF, 10 V, 0805/1206), or a polymer tantalum or aluminium
   capacitor (check its leakage against the about 1 mA idle current). The BLE battery reads, which
   sample at the dip bottom, would then read close to the true bank voltage.
   Assumptions: Rs is the bank ESR plus wiring and does not change with voltage; the radio events
   during the hold-up carry the same charge as the idle events of test 1.